

Problem 3

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Problem 1 Give an example or sketch of a function that is continuous on $(-\infty, \infty)$ and satisfies given properties. If such a function does not exist, explain why.

- (a) A function f is concave up and negative everywhere.
- (b) A function f is decreasing and concave up everywhere.
- (c) A function s has exactly 3 local extrema and four inflection points.
- (d) A function f has exactly 2 zeros and one local extrema.